```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, norm
n = 10
p = 0.5
x = np.arange(0, n + 1)
prob = binom.pmf(x, n, p)

print("Binomial Probabilities:")
print(prob)
data = np.linspace(-4, 4, 100)
normal_prob = norm.pdf(data, 0, 1)
P_A = 0.01
P_B_given_A = 0.95
P_B_given_not_A = 0.05

P_B = (P_B_given_A * P_A) + (
    P_B_given_not_A * (1 - P_A)
)

P_A_given_B = (P_B_given_A * P_A) / P_B

print("\nBayes' Theorem:")
print("P(A|B) =", round(P_A_given_B, 4))

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.bar(x, prob)
plt.title("Binomial Distribution")
plt.xlabel("Number of Successes")
plt.ylabel("Probability")
plt.subplot(1, 2, 2)
plt.plot(data, normal_prob)
plt.title("Normal Distribution")
plt.xlabel("Value")
plt.ylabel("Probability Density")
plt.tight_layout()
plt.show()
```

